Your grade: 98%

Your latest: **98%** • Your highest: **98%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following best describes the role of AI in the expression "an AI-powered society"? 1 / 1 point

- ○ AI helps to create a more efficient way of producing energy to power industries and personal devices.
- ○ AI controls the power grids for energy distribution, so all the power needed for industry and in daily life comes from AI.
- ● AI is an essential ingredient in realizing tasks, in industry and in personal life.

Nice work

In an AI-powered society AI plays a fundamental role to complete most tasks, in industry and personal life.

2. Which of the following play a major role to achieve a very high level of performance with Deep Learning algorithms? 0.8 / 1 point

- [ ] Smaller models.
- [ ] Better designed features to use.
- [x] Large models.

Nice work

Yes. In most cases it is necessary for a very large neural network to make use of all the available data.

- [x] Large amounts of data.

Nice work

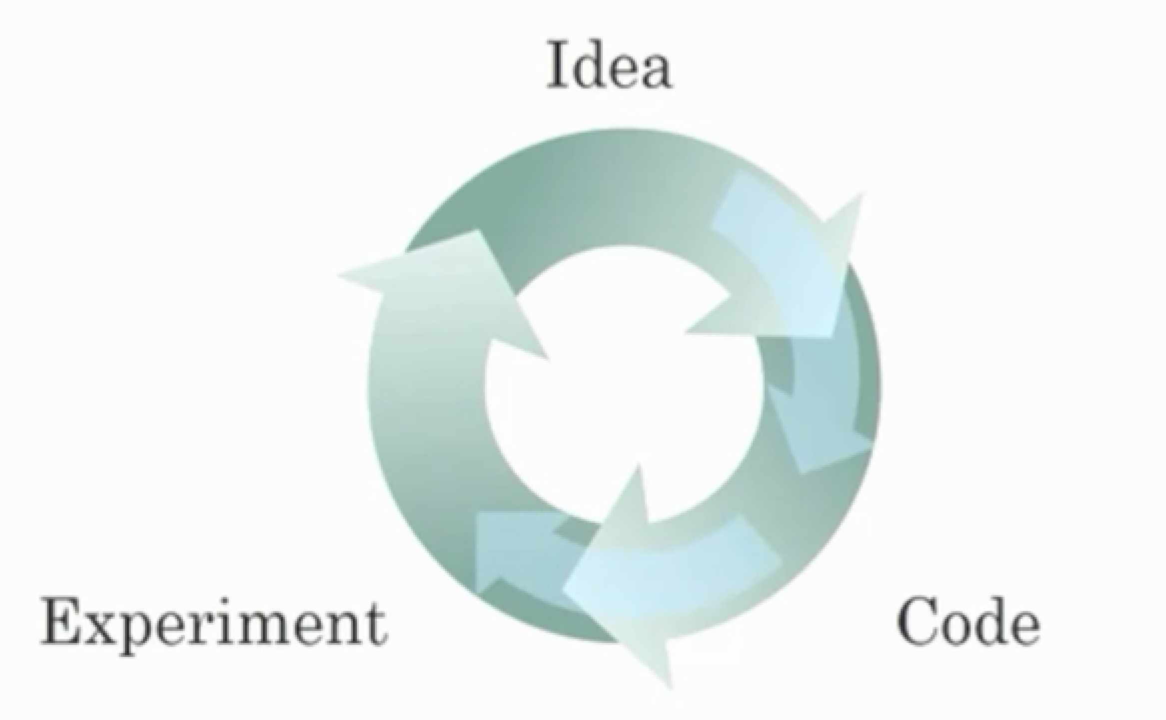
Yes. Some of the most successful Deep Learning algorithms make use of very large datasets for training.

- [ ] Deep learning has resulted in significant improvements in important applications such as online advertising, speech recognition, and image recognition.

You didn't select all the correct answers

3. Recall this diagram of iterating over different ML ideas. Which of the statements below are true? (Check all 1 / 1 point

3. **Recall this diagram of iterating over different ML ideas. Which of the statements below are true? (Check all that apply.)** 1 / 1 point



- [x] Better algorithms can speed up the iterative process by reducing the necessary computation time.

  Nice work

  Yes. Recall how the introduction of the ReLU activation function helped reduce the time needed to train a model.

- [x] Improvements in the GPU/CPU hardware enable the discovery of better Deep Learning algorithms.

  Nice work

  Yes. By speeding up the iterative process, better hardware allows researchers to discover better algorithms.

- [ ] Better algorithms allow engineers to get more data and then produce better Deep Learning models.
- [ ] Larger amounts of data allow researchers to try more ideas and then produce better algorithms in less time.

4. **Neural networks are good at figuring out functions relating an input x to an output y given enough examples. True/False?** 1 / 1 point

4. Neural networks are good at figuring out functions relating an input x to an output y given enough examples. True/False?

1 / 1 point

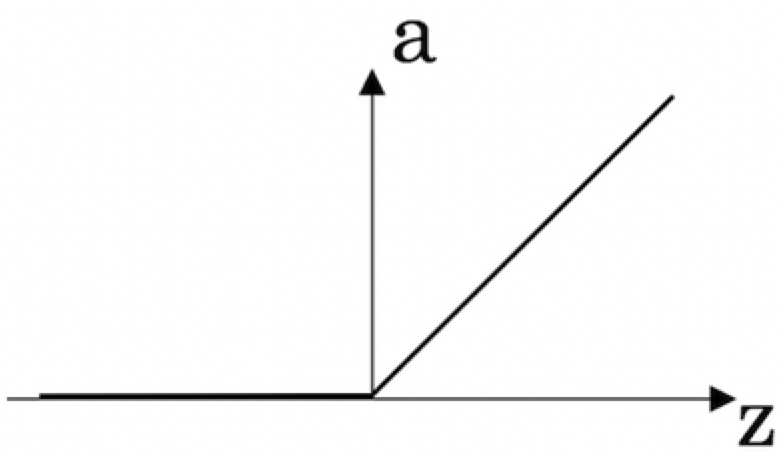
- False
- True

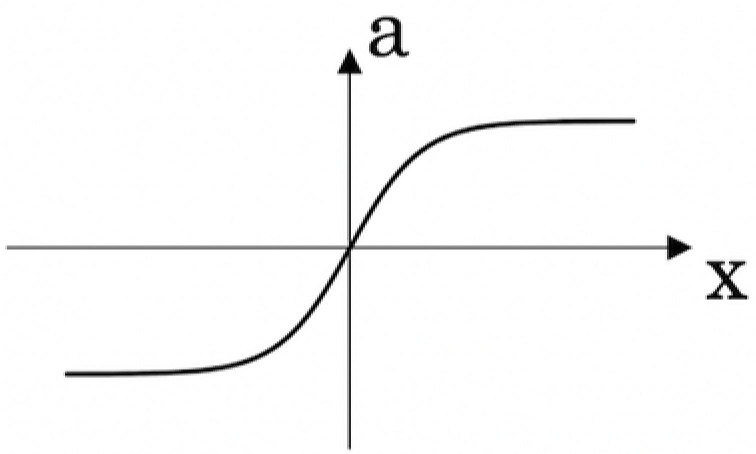
Nice work

Exactly, with neural networks, we don't need to "design" features by ourselves. The neural network figures out the necessary relations given enough data.

5. Which of the following depicts a Sigmoid activation function?

1 / 1 point

- 

- 

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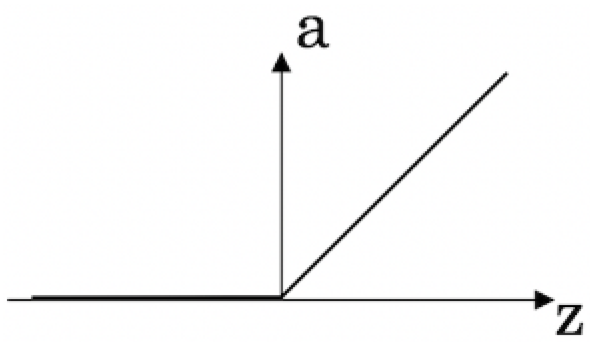
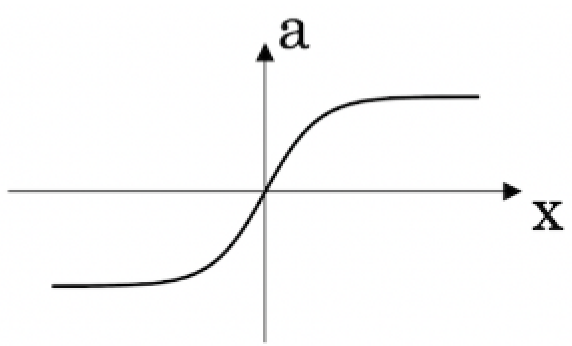
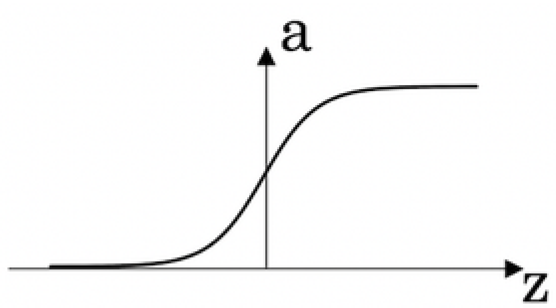
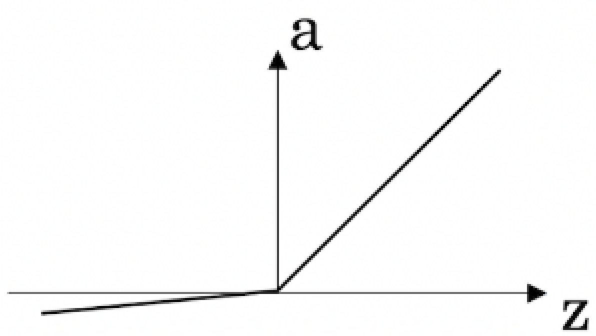
5. Which of the following depicts a Sigmoid activation function?

1 / 1 point

- [ ] 

- [ ] 

- [x] 

- [ ] 

Nice work

Correct! This is the sigmoid activation function; this function was changed for the ReLU activation function helping with the training of NN.

Nice work

Correct! This is the sigmoid activation function; this function was changed for the ReLU activation function helping with the training of NN.

6. **Features of animals, such as weight, height, and color, are used for classification between cats, dogs, or others. This is an example of "structured" data, because they are represented as arrays in a computer. True/False?** 1 / 1 point

- ( ) False

  No. The data can be represented by columns of data. This is an example of structured data, unlike images of the animal.

- (•) True

  Yes. The data can be represented by columns of data. This is an example of structured data, unlike images of the animal.

Nice work

7. **Which of the following are examples of structured data? Choose all that apply.** 1 / 1 point

- [ ] A dataset with short poems.
- [x] A dataset with zip code, income, and name of a person.

  Nice work

  Yes, this data can be presented in a table. This is an example of "structured" data.

- [x] A dataset of weight, height, age, the sugar level in the blood, and arterial pressure.

  Nice work

  Yes, this data can be presented in a table. This is an example of "structured" data.

- [ ] A set of audio recordings of a person saying a single word.

8. **Why can an RNN (Recurrent Neural Network) be used to create English captions to French movies? Choose all that apply.** 1 / 1 point

- [ ] The RNN requires a small number of examples.

- [ ] A set of audio recordings of a person saying a single word.

8. Why can an RNN (Recurrent Neural Network) be used to create English captions to French movies? Choose all that apply.

1 / 1 point

- [ ] The RNN requires a small number of examples.
- [x] The RNN is applicable since the input and output of the problem are sequences.

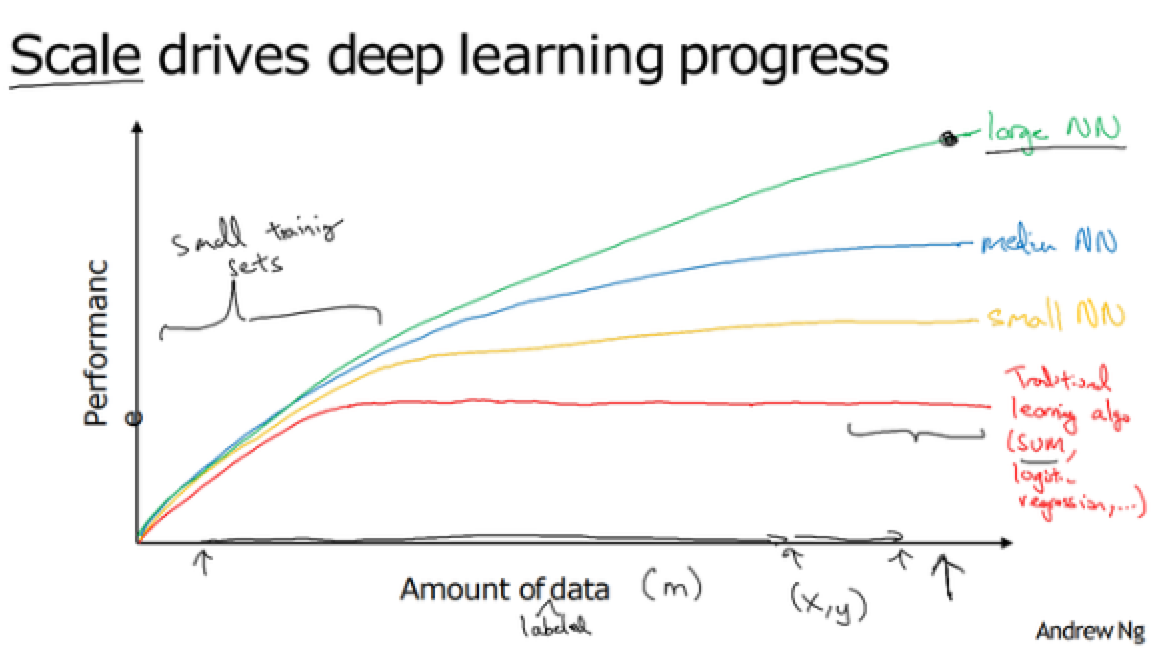
> **Nice work**
>
> Yes, an RNN can map from a sequence of sounds (or audio files) to a sequence of words (the caption).

- [x] It can be trained as a supervised learning problem.

> **Nice work**
>
> Yes, the data can be used as x (movie audio) to y (caption text).

- [ ] RNNs are much more powerful than a Convolutional neural Network (CNN).

1 / 1 point



9. From the given diagram, we can deduce that Large NN models are always better than traditional learning algorithms. True/False?

- (•) False
- ( ) True

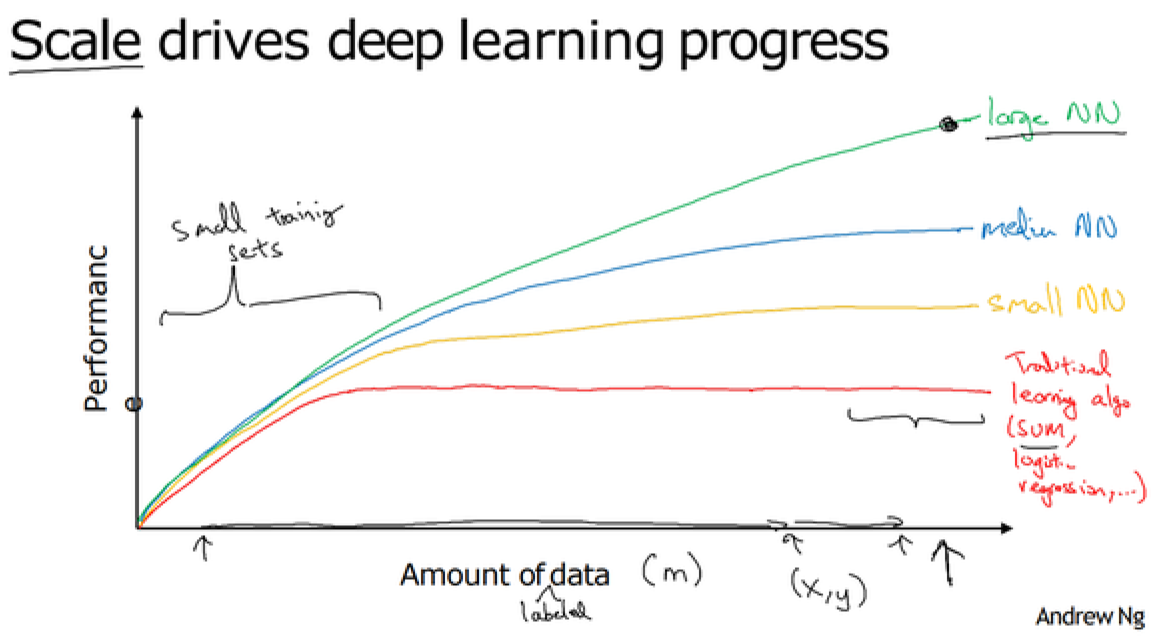
9. **From the given diagram, we can deduce that Large NN models are always better than traditional learning algorithms. True/False?**

- (•) False
- ( ) True

Nice work

Yes, when the amount of data is not large the performance of traditional learning algorithms is shown to be the same as NN.

10. **Assuming the trends described in the figure are accurate. The performance of a NN depends only on the size of the NN. True/False?** 1 / 1 point



- ( ) True
- (•) False

Nice work

Yes. According to the trends in the figure above, It also depends on the amount of data.